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# CSMS role in product development stages

## IVI LD1 – Cyber Security Part

# 1. UNECE WP29 R155 : Cybersecurity Regulation

## Overview:

- As cybersecurity has emerged as an important criterion for automotive safety, the United Nations Economic Commission for Europe (UNECE) International Harmonization In June 2020, the UN Regulation No.155: Cybersecurity Regulation was adopted and came into effect on January 21.
- Applicable to new vehicles from July 2022 and all vehicles from July 2024 for countries subject to EU laws

## Highlights:

❑ **Establish a CSMS (Cybersecurity Management System) and apply cybersecurity to vehicles in accordance with the CSMS.**

- [OEM] Acquisition of CSMS Certificate - Acquisition of > Vehicle VTA
- [Supplier] CSMS Implementation-> Support for OEM to acquire VTA (Evidence Provided)

❑ **Certificate of Compliance for CSMS (Process perspective)**

- Cybersecurity process certification for your cybersecurity organization, R&R, and vehicle lifecycle
- Design Perspective: Identify and manage cyber security risks
- Verify that identified risks are managed through testing, etc.
- Ongoing management of risk assessments
- Monitor and respond to cyberattacks
- Support analysis of all attacks (attempted or completed)
- Assess whether cybersecurity measures are effective against new threats and vulnerabilities

❑ **Vehicle Type Approval (Actual Vehicle Execution/Application Viewpoint)**

- Application of security technology by vehicle model and certification of evaluation results (actual vehicle, functional test)
- Evidence that you have a Cybersecurity Management System (CSMS) in place and that it has been applied to your vehicle.
- Analyze the results of the risk assessment and identify priorities (Critical)
- Mitigation to reduce cybersecurity risks
- Verification through cybersecurity testing to prove that countermeasures are working
- Action Points to Detect and Prevent Cyberattacks (IDPS)
- Security logger
- Appropriate monitoring for vehicle type and reporting monitoring activities to the approval body

## OEM/Tier1 relations for UNECE WP29 R155 correspondence

**OEM**

- Establishment of CSMS system
- In accordance with the CSMS system Cybersecurity in vehicles application
- Acquired CSMS certification
- VTA (Vehicle Type Approval)

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**Supplier**

- Establishment of CSMS system
- In accordance with the CSMS system Products in Cybersecurity application
- VTA acquisition point of OEM Evidence for the Circle offer

## UNECE WP29 R155 and ISO/SAE 21434 Relationship

Regulation **UNECE WP29 R155** → Standard **ISO/SAE 21434**

❑ Requires CSMS at high level

- (7.2) Requirements for the Cyber Security Management System
- (7.2.1) For the assessment the Approval Authority or its Technical Service shall verify that the vehicle manufacturer has a Cyber Security Management System in place and shall verify its compliance with this Regulation.

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❑ Provides guideline how to implement a CSMS

- (Clause 5) Organizational cybersecurity management
- (Clause 6) Project dependent cybersecurity management
- (Clause 7) Distributed cybersecurity activities
- (Clause 8) Continuous cybersecurity activities

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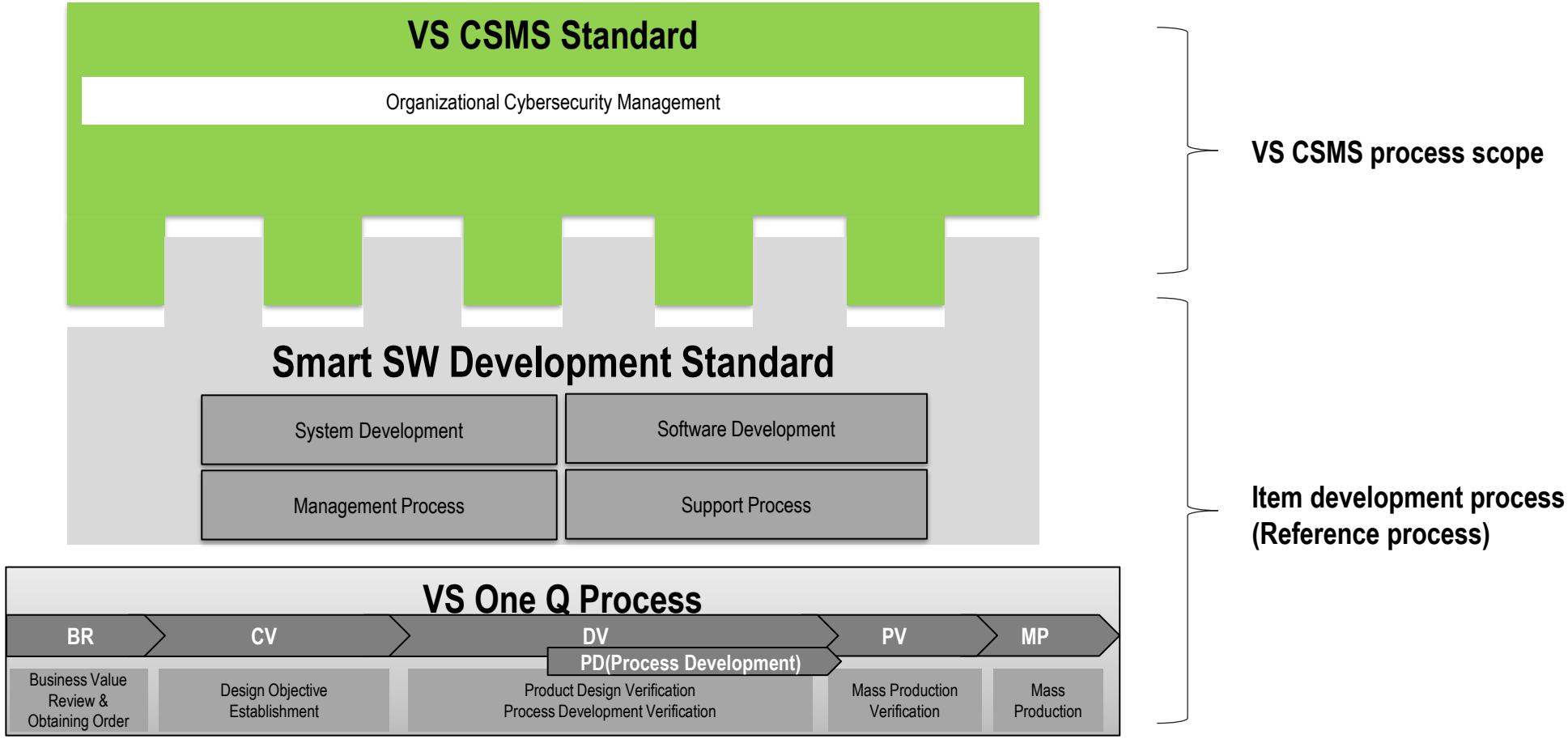
- ISO/SAE 21434 supports to implement a CSMS as it is required by
- UNECE R155 ISO/SAE 21434 only describes the tasks for a process, but leaves the design to the users

# 2. VS CSMS standard composition

- The scope of the VS CSMS standard is limited to the cybersecurity area based on UNECE Cybersecurity Regulation, ISO/SAE 21434 and LGE Regulation of the SW Development Standard Process.
- To enable cybersecurity engineering, an organization institutes and maintains cybersecurity governance and cybersecurity culture. This involves specifying organization-specific rules and processes covering concept, development, production, operation, maintenance and decommissioning, including cybersecurity risk management, information sharing, vulnerability disclosure, cybersecurity monitoring, and incident response.
- Reference process applies the development process used by VS business unit.



# 3. CSMS activity definition

## Organizational Cybersecurity Management

**Organizational Cybersecurity Management**

- Cybersecurity Governance
- Cybersecurity Culture
- Information Sharing
- Management System
- Tool Management
- Information Security Management
- Organizational Cybersecurity Audit

## Cybersecurity Concept Definition

**Definition of Cybersecurity goals**

- Item Definition
- Threat analysis and risk assessment(TARA)
- Cybersecurity Goals specification

**Definition of Cybersecurity requirements**

- Cybersecurity Requirements specification
- Obtaining Cybersecurity requirements
- Review Cybersecurity requirements

## Cybersecurity System Development

**Cybersecurity System design**

- Initiation of system development
- (system)Cybersecurity requirements specification
- Release of (system)cybersecurity requirements
- System (cybersecurity) architectural design
- Release of System (cybersecurity) architectural design
- Cybersecurity System Integration test specification
- Cybersecurity System Qualification test specification

**Cybersecurity System Test & Validation**

- Cybersecurity System Integration Test
- Cybersecurity System Qualification Test
- (System) Penetration Test
- Vehicle Validation
- System Release

## Cybersecurity HW Development

**HW Cybersecurity Design**

- Initiation of HW development
- (HW) cybersecurity requirements specification
- Release of (HW) cybersecurity requirements
- HW (cybersecurity) architecture design
- HW (cybersecurity) design release
- Cybersecurity HW Qualification Test Specification

**HW Cybersecurity Verification**

- Cybersecurity HW Integration Test
- Cybersecurity HW Qualification Test
- HW release

## Cybersecurity SW Development

**SW Cybersecurity Design**

- Initiation of SW development
- (SW)cybersecurity requirement specification
- Release of (SW)cybersecurity requirement
- SW (cybersecurity) architecture design
- Release of SW (cybersecurity) architecture
- SW unit cybersecurity design and implementation
- Cybersecurity SW Qualification Test Specification

**SW Cybersecurity Verification**

- Cybersecurity SW Unit Test
- Cybersecurity SW Integration Test
- Cybersecurity SW Qualification Test
- Cybersecurity Vulnerability Test
- SW Release

## Cybersecurity Management & Supporting

**Cybersecurity Management**

- Establish Release Cybersecurity Plan
- Development of CIA & suppliers' CIA
- Release of CIA of Cybersecurity Plan
- Establish Cybersecurity Audit
- Establish Cybersecurity Assessment

**Cybersecurity Gate Review(Q-Gate)**

- Requirements confirm review
- Feature Complete Review
- Qualification completion Review

**Supporting**

- Configuration Management
- Requirement Change Management
- Vulnerabilities Change Management

## Production & Operation for Cybersecurity

**Cybersecurity production**

- Production control plan

**Continuous Cybersecurity Management**

- Cybersecurity threat monitoring
- Initial incident analysis
- Confirm affected products
- Detailed threat analysis
- Threat countermeasure development management
- Prepare threat countermeasure
- Post-Response Activities
- End of Cybersecurity Support

**Cybersecurity Case**

- Planning of Cybersecurity Case
- Development of Cybersecurity Case

# 4. Steps in each product development stage

| | DV | PV | Pre-MP | MP |
|---|---|---|---|---|
| **LGE** | | | | |
| **Objectives** | Acquire Production Cybersecurity Requirements | Prepare cybersecurity documentations In compliance with requirements | | Continue to update documentations from PV and perform assessment on Tier-2 |
| **CSM** | **Product Design and Development**<br>1. Production Control Plan (http://collab.lge.com/main/x/rOdzrQ)<br>2. Production Cybersecurity Requirements (http://collab.lge.com/main/pages/viewpage.action?pageId=3184188188) | **Product and Process Validation**<br>3. Feasibility checking report of cybersecurity requirements. ([VSCSGU-4489] No.1 Process & Checklist - NEXT JIRA)<br>4. Applicability assessment report ([VSCSGU-4491] No.3 Process & Checklist - NEXT JIRA)<br>5. Archive assessment and inspection results ([VSCSGU-4492] No.4 Process & Checklist - NEXT JIRA)<br>6. SW Tools management process each project ([VSCSGU-4493] No.5 Process & Checklist - NEXT JIRA)<br>7. Security Tools Management category risk assessment report ([VSCSGU-4494] No.6 Process & Checklist - NEXT JIRA)<br>8. Manage cybersecurity competency tracking<br>9. Risk assessment of management processes (key management, transfer EDI, rework, … processes) [VSCSGU-4495] No.7 Process & Checklist - NEXT JIRA | | **Mass Production**<br>10.Perform vulnerability monitoring and respond incident response activity ([VSCSGU-4500] No.12 Process & Checklist - NEXT JIRA)<br>11. Continue to collect assessment and inspection results<br>12. Follow buyer audit and prepare documents.<br>13.Conduct internal audit of each project<br>14. Monitor improvement and vulnerability remediation activities<br>15.Perform assessment on Tier-2 ([VSCSGU-4501] No.13 Process & Checklist - NEXT JIRA) |
| **CSG** | 1. Provide regular training session ([VSCSGU-4498] No.10 Process & Checklist - NEXT JIRA)<br>2. Provide process training session ([VSCSGU-4495] No.7 Process & Checklist - NEXT JIRA)<br>3. Manage and archive education completion status & cybersecurity capability status of employees.([VSCSGU-4499] No.11 Process & Checklist - NEXT JIRA)<br>4. Establish and maintain processes including TARA, vulnerability management, incident response, vendor management for all projects<br>5. Collaborate with IT/security team to verify physical access security, information security and facility security ([VSCSGU-4496] No.8 Process & Checklist - NEXT JIRA)<br>6. Monitor IT information security certificates ([VSCSGU-4497] No.9 Process & Checklist - NEXT JIRA)<br>7. Archives assessment result and inspection results as evidence for VTA ([VSCSGU-4492] No.4 Process & Checklist - NEXT JIRA)<br>8. Audit preparation and reporting | | | |

DV (Design Verification), IQ (Incoming Quality)
PV (Product Verification), OQ (Outgoing Quality)
TARA: Threat Analysis and Risk Assessment, VTA: Vehicle Type Approval

# 5. Main role in product development stage

| No | Role | Phase | Template |
|---|---|---|---|
| 1 | Security part reviews production cybersecurity requirements and conduct feasibility checking whether the requirements can be applied and inspected in production line | PV | Feasibility<br>ing report of cybe |
| 2 | Security part applies production cybersecurity requirements into production control plan and maintain the production control plan | | |
| 3 | Security part performs assessment on inspection results to verify that the product is manufactured which is complying to production cybersecurity requirements in production control plan.<br>(Ex. correct approved SW firmware/version shall be injected into product, correct keys shall be injected into product, removal of JTAG/Debug connector, secure boot enabled, sealed HW plug ,etc) | | Applicability<br>assessment report |
| 4 | Security part archives assessment result and inspection results as evidence for VTA | | Archive<br>nent and inspectio |
| 5 | Security part manages inventory of tools and equipment for (secure) production.<br>(Ex. flashing software, test tools, ROM writing & Inspection tools such as Data management Software(DMS) / Tasklink / Flashcore / Lumenx FA & FCT Inspection Program) | | SW Tools<br>ment process each |
| 6 | Security part performs risk assessment regarding assets such as tools & equipment for (secure) production which shall be tool version management, obtaining rationale that tools are managed by strong access control and etc. | | Security Tools<br>agement category |
| 7 | Security part understands all process from start of production to end of production (OQC) and obtains evidence with rationale in perspective of cybersecurity of every step<br>(ROM Writing ~ key injection ~ inspection ~ OQC ~ just before shipping) | | Risk assessment<br>nanagement proce |
| 8 | Security part collaborate with IT/security team to verify physical access security, information security and facility security. | All | CSG will implement |
| 9 | Security part monitor IT information security certificates such as TISAX, ISO 27001, IATF16949, etc | | |
| 10 | Security part provides regular training session to employees, publish cybersecurity newsletter by translating to VietNam language of HQ newsletter and runs a cybersecurity campaign to enhance cybersecurity culture. | | |
| 11 | Security part manages and archives education completion status & cybersecurity capability status of employees. | | |
| 12 | Security part performs vulnerability monitoring on production line/system/tools and responds it as incident response activity. | MP | CSM continue to update documentations from PV and perform assessment on Tier-2 |
| 13 | Security part performs assessment on Tier-2 (tool & equipment company, outsourced production company) | | |

# 5. Main role in product development stage

## Feasibility checking report of cybersecurity requirements.

| Step | Input | Department / Team | | | Output | Lead time |
|---|---|---|---|---|---|---|
| | | CSM | HQ ME | Relevant Teams (HQ SWPL, CSM & VH ME, PD, LD1) | | |
| Acquire Document | •Cybersecurity requirements<br>•Production control plan | 1 | Provide production control plan including production cybersecurity requirements | | •Cybersecurity requirements<br>•Production control plan | At the end of DV phase |
| Review Document | •Customer requirement<br>•Production control plan<br>•Internal requirement | 2<br>Analysis requirements | | | Checklist items | D+3 |
| Feasibility checking | Checklist items | 3<br>With relevant teams feasibility checking<br>OK<br>Confirm cybersecurity requirements in production control plan | NG | Review & analysis regarding feasibility | Assessment report | D+5 |
| Release final document | Assessment report | 4 | | Release final production control plan | Production control plan | D+7 |

## Applicability assessment report

| Step | Input | Department / Team | | | Output | Lead time |
|---|---|---|---|---|---|---|
| | | CSM | VH ME | Relevant Teams (VH PD, ETT, LD1) | | |
| Acquire Document | Production control plan | 1 | Provide final production control plan including production cybersecurity requirements | | HW, SW & tool requirements items | At final production control plan released |
| Review Document | HW, SW & tool requirements items | 2 Determine execution teams | | | Checklist items | D+3 |
| Feasibility checking | Checklist items | 3 Applicability assessment (NG / OK) | | Execute cybersecurity requirements into production process<br>Provide inspection result | Assessment report | D+30<br>D+30<br>D+30 |
| Release final document | Assessment report | 4 Assessment report | | | Applicability assessment report | D+30 |

## Archive assessment and inspection results

| Step | Input | Department / Team | | Output | Lead time |
|---|---|---|---|---|---|
| | | CSM | CSG | | |
| Acquire Document | • Assessment report<br>• Inspection result | 1 Provide cybersecurity assessment report and inspection result | | • Assessment report<br>• Inspection result | At applicability assessment report released |
| Review Document | • Assessment report<br>• Inspection result | 2 | Collect documents | Summary file | D+3 |
| Feasibility checking | Summary file | 3 Review and update assessment report and inspection result | NG<br>Accuracy assessment<br>OK<br>Confirm the accuracy of documents | Assessment report | D+7 |
| Release final document | Assessment report | 4 | Document archiving | Evidence documents for VTA | D+7 |

## SW Tools management process each project

| Step | Input | Department / Team | | | Output | Lead time |
|---|---|---|---|---|---|---|
| | | VH CSM | VH ME | Relevant Teams (VH ETT, PD, LD1) | | |
| Acquire Document | • Production control plan<br>• Tool versions list | 1 | Provide final production control plan including tool versions | Provide latest tool versions | | At final production control plan released |
| Review Document | • Production control plan<br>• Tool versions list | 2<br>Collect tool version management file | | | Summary file | D+3 |
| Feasibility checking | Summary file | 3<br>Cross-check tool version information<br>OK<br>Confirm final tool versions | Update correct versions<br>NG | | Assessment report | D+5<br>D+5<br>D+5 |
| Release final document | Assessment report | 4<br>Archive final tool versions result | | | Tool version management report | D+5 |

## Security Tools Management category risk assessment report

| Step | Input | Department / Team | | Output | Lead time |
|---|---|---|---|---|---|
| | | VH CSM | Relevant Teams (VH ETT, PD, LD1) | | |
| Acquire Document | • Tool version management report<br>• Tools and equipment information | 1 | Provide identified assets such as tools and equipment information in production line | | |
| Review Document | • Tool version management report<br>• Tools and equipment information | 2<br>Create a damage scenario | | Damage scenario list | D+5 |
| Analysis and assessment | Damage scenario list | 3<br>Conduct impact rating based on S-F-O-P values<br>Conduct threat scenario identification<br>Attack path analysis<br>Attack feasibility rating<br>Risk value determination | | • Impact rating result<br>• Threat scenario list<br>• Attack feasibility rating result | D+7 |
| Release final document | | 4<br>Risk treatment decision | | Threat analysis and risk assessment report | D+7 |

## Risk assessment of management processes

| Step | Input | Department / Team | | | Output | Lead time |
|---|---|---|---|---|---|---|
| | | VH CSM | VH ME | Relevant Teams (VH ETT, PD, LD1, etc) | | |
| Acquire Document | • Production control plan<br>• Production route chart | 1 | Provide production processes in production control plan | Provide processes map | • Processes map | |
| Review Document | • Processes map | 2<br>Analyze and identify processes for cybersecurity compliance and implementation | | | • Evidence list | D+5 |
| Analysis and assessment | • Evidence list | 3<br>Collect evidence of each process<br>Conduct completeness and accuracy assessment<br>Enough<br>Confirm evidence of each process | Review and update missing content<br>Not enough | | • Evidence assessment result | D+5 |
| Release final document | | 4<br>Archive evidence of each process | | | Evidence of each process report | D+5 |

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# Thank you